Wave motion (Supplementary Problems)

1. The average person can hear sound waves ranging in frequency from about 20 Hz to 20 kHz. Determine the wavelengths at these limits, taking the speed of sound to be 340 m/s. Ans. 17 m, 1.7 cm

2. Radio station WJR in Detroit broadcasts at 760 kHz. The speed of radio waves is 3.00×10^8 m/s. What is the wavelength of WJR's waves? Ans. 395 m

3. Radar waves with 3.4 cm wavelength are sent out from a transmitter. Their speed is 3.00×10^8 m/s. What is their frequency? Ans. 8.8 × 10 Hz = 8.8 GHz

4. When driven by a 120 Hz vibrator, a string has transverse waves of 31 cm wavelength traveling along it. (a) What is the speed of the waves on the string? (b) If the tension in the string is 1.20 N, what is the mass of 50 cm of the string? Ans. (a) 37 m/s; (b) 0.43 g

5. The wave shown in Fig. 22-9 is being sent out by a 60 cycle/s vibrator. Find the following for the wave: (a) amplitude, (b) frequency, (c) wavelength, (d) speed, (e) period. Ans. (a) 3.0 mm; (b) 60 Hz; (c) 2.00 cm; (d) 1.2 m/s; (e) 0.017 s

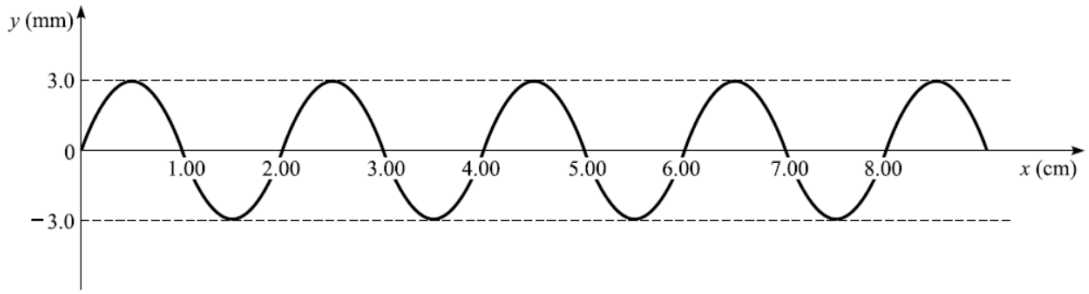


Fig. 22-9

6. A copper wire 2.4 mm in diameter is 3.0 m long and is used to suspend a 2.0 kg mass from a beam. If a transverse disturbance is sent along the wire by striking it lightly with a pencil, how fast will the disturbance travel? The density of copper is 8920 kg/m3. Ans. 22 m/s

7. A string 180 cm long resonates in three segments to transverse waves sent down it by a 270 Hz vibrator. What is the speed of the waves on the string? Ans. 324 m/s

8. A string resonates in three segments to a frequency of 165 Hz. What frequency must be used if it is to resonate in four segments? Ans. 220 Hz

9. A flexible cable, 30 m long and weighing 70 N, is stretched between two poles by a force of 2.0 kN. If the cable is struck sideways at one end, how long will it take the transverse wave to travel to the other end and return? Ans. 0.65 s

10. A wire under tension vibrates with a fundamental frequency of 256 Hz. What would be the fundamental frequency if the wire were half as long, twice as thick, and under one-fourth the tension? Ans. 128 Hz

11. Steel and silver wires of the same diameter and same length are stretched with equal tension. Their densities are 7.80 g/cm^3 and 10.6 g/cm^3, respectively. What is the fundamental frequency of the silver wire if that of the steel is 200 Hz? Ans. 172 Hz

12. A string has a mass of 3.0 gram and a length of 60 cm. What must be the tension so that when vibrating transversely its first overtone has frequency 200 Hz? Ans. 72 N

13. (a) At what point should a stretched string be plucked to make its fundamental tone most prominent? At what point should it be plucked and then at what point touched (b) to make its first overtone most prominent and (c) to make its second overtone most prominent? Ans. (a) center; (b) plucked at 1/4 of its length from one end, then touched at center; (c) plucked at 1/6 of its length from one end, then touched at 1/3 of its length from that end

14. What must be the length of an iron rod that has the fundamental frequency 320 Hz when clamped at its center? Assume longitudinal vibration at a speed of 5.00 km/s. Ans. 7.81 m

15. A rod 120 cm long is clamped at the center and is stroked in such a way as to give its first overtone. Make a drawing showing the location of the nodes and antinodes, and determine at what other points the rod might be clamped and still emit the same tone. Ans. 20.0 cm from either end

16. A metal bar 6.0 m long, clamped at its center and vibrating longitudinally in such a manner that it gives its first overtone, vibrates in unison with a tuning fork marked 1200 vibration/s. Compute the speed of sound in the metal. Ans. 4.8 km/s

17. Determine the length of the shortest air column in a cylindrical jar that will strongly reinforce the sound of a tuning fork having a vibration rate of 512 Hz. Use n 340 m/s for the speed of sound in air. Ans 16.6 cm

18. A long, narrow pipe closed at one end does not resonate to a tuning fork having a frequency of 300 Hz until the length of the air column reaches 28 cm. (a) What is the speed of sound in air at the existing room temperature? (b) What is the next length of column that will resonate to the fork? Ans. (a) 0.34 km/s; (b) 84 cm 22.37 An organ pipe closed at one end is 61.0 cm long. What are the frequencies of the first three overtones if n for sound is 342 m/s? Ans. 420 Hz, 700 Hz, 980 Hz

19. A wave on a Slinky has a frequency of 2.3 Hz. It takes 3.5 s to travel the length of the Slinky (0.30 m). What is the wavelength?

20. Sound waves in water travel at a speed of approximately 1500 m/s. What is the wavelength of a sound wave that has a frequency of 440 Hz?

21. SSM A wave on a string is described by the relation $y = A \sin(35t - 0.025x)$, where t is measured in seconds and x in meters, with $A = 0.15$ m. Find the frequency, wavelength, and speed of this wave.

22. A wave on a string is described by the relation $y = A \sin(72t - 0.045x)$, where t is measured in seconds and x in meters, with $A = 0.15$ m. What is the maximum speed of a point on the string?

23. Using Equation 12.1 $y = A\sin\left(2\pi ft - \frac{2\pi x}{\lambda}\right)$, write the equation of a wave on string that has an amplitude of 0.30 m, a frequency of 350 Hz, and a velocity of 250 m/s and travels to the right.

24. Using Equation 12.1, write the equation of a wave on string that has an amplitude of 0.50 m, a wavelength of 8.4 m, and a velocity of 330 m/s and travels to the left.

25. Consider a wave that has a frequency of 3.0 kHz and a wavelength of 0.10 m. Write an equation that might describe this

wave. 15. Figure P12.15 shows a snapshot of a wave on a string. This wave has a frequency of 200 Hz. Approximately how long does it take the wave crest located at point A in this snapshot to move to point B?

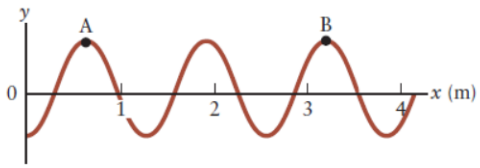


Figure P12.15

26. Draw a snapshot of a wave on a string that obeys the equation y = (4.2 mm)cos(3000t - 0.45x). Assume the units of length are millimeters and the units of time are seconds. Make your drawing at t = 1.5 s.

27. A wave on a string is described by the equation y = (4.2 mm) cos (3000t - 0.45x) where t is measured in s and x is measured in mm. What are the frequency and wavelength of the wave?

28. A wave on a string is described by the equation y = (4.2 mm) cos(3000t - 0.45x) where t is measured in s and x is measured in mm. Make a graph of the displacement of the string at x = 0.35 mm as a function of time. Show two complete periods of the wave.

29. Consider a wave that has a frequency of 3.0 kHz and a wavelength of 0.10 m. What is the speed of this wave?

30. Two sound waves in air have frequencies that differ by a factor of fi ve. What is the ratio of their wavelengths?

31. A wave on a string has a speed of 340 m/s. If the tension in the string is 150 N and its length is 2.0 m, what is the mass of the string?

32. Consider a piano string made of steel with a diameter of 1.0 mm. If the speed of a wave on this string is 280 m/s, what is the tension in the string? Hint: The density of steel is 7800 kg/m^3.

33. Waves on a violin string travel at 900 m/s. If the string has a tension of 50 N and is 0.33 m long, what is its mass?

34. A wave on a string has a speed of 200 m/s. If the diameter of the string is increased by a factor of two and the tension is held fixed, what is the speed of the wave on this new string?

35. SSM Consider a violin string with a vibrating length of 30 cm. If the speed of a wave on this string is 250 m/s, what is the tension in the string? Hint: You will have to estimate the diameter of the string. Assume the density is the same as that of

steel, 7800 kg/m^3.

36. The speed of a wave on a guitar string is v = 120 m/s. If the tension is increased from 70 N to 110 N, what is the new value of v?

37. The speed of sound in water is 1500 m/s. Find the wavelength of a sound wave whose frequency is 300 Hz.

38. Spruce is a type of wood that is widely used in making musical instruments (such as violins and guitars) because its

39. Suppose your favorite FM radio station broadcasts at a frequency of 101 MHz. What is the wavelength of this signal?

40. Your favorite television station transmits using an electromagnetic wave with a wavelength of approximately 1.5 m. What is the frequency?

41. A violin string is 35 cm long and has a fundamental standing wave frequency of 440 Hz. If the tension is 75 N, what is the mass of the string?

42. A vibrating string has a fundamental frequency of 250 Hz. The tension is then increased by a factor of 1.9. What is the new value of the fundamental frequency?

43. A violin string has an initial tension of 45 N. When playing the note A on the violin, the frequency is found to be 435 Hz. This note should actually have the frequency 440 Hz, so the player increases the tension to get the correct frequency. What is the new value of the tension? Give your answer to three significant figures.

44. A steel piano string for the note middle C is about 0.60 m long and is under a tension of 600 N. If the fundamental frequency is 262 Hz, what is the string's diameter?

45. The vibrating length of a guitar string is reduced by a factor of 2.5 when a player holds the string against a fret. What is the change in the frequency of the fundamental standing wave on the string?

46. Two nearby nodes of a standing wave are separated by 3.5 m. If the frequency of the wave is 150 Hz, what is the speed of the wave?

47. Consider a standing wave that has a frequency of 2000 Hz and a speed of 300 m/s. What is the distance between a node and the nearest antinode?

48. Figure P12.56 shows a standing wave on a string. If the frequency of the wave is 440 Hz, what is the speed of a wave on this string?

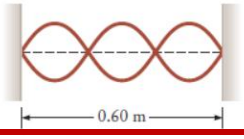


Figure P12.56

49. You want to set up a standing wave on a string that has a length of 3.5 m. You find that the lowest frequency that will work is 20 Hz. What is the speed of a wave on this string?

50. A sinusoidal wave is traveling along a rope. The oscillator that generates the wave completes 40.0 vibrations in 30.0 s. A given crest of the wave travels 425 cm along the rope in 10.0 s. What is the wavelength of the wave?

51. These two waves travel along the same string: $y_1(x, t) = (4.60\ mm) \sin(2\pi x - 400\pi t)$, $y_2(x, t) = (5.60\ mm) \sin(2\pi x - 400\pi t + 0.80\pi\ rad)$. What are (a) the amplitude and (b) the phase angle (relative to wave 1) of the resultant wave? (c) If a third wave of amplitude 5.00 mm is also to be sent along the string in the same direction as the first two waves, what should be its phase angle in order to maximize the amplitude of the new resultant wave?

52. A string oscillates according to the equation : $y = (0.50\,cm)\sin\left[\left(\frac{\pi}{3}cm^{-1}\right)x\right]\cos\left[\left(40\pi s^{-1}\right)t\right]$

What are the (a) amplitude and (b) speed of the two waves (identical except for direction of travel) whose superposition gives this oscillation? (c) What is the distance between nodes? (d) What is the transverse speed of a particle of the string at

the position x = 1.5 cm when t = 9/8 s?

53. The equation of a transverse wave traveling along a string is

$$y = (2.0\,mm)\sin\left[\left(20m^{-1}\right)x - \left(600s^{-1}\right)t\right]$$

Find the (a) amplitude, (b) frequency, (c) velocity (including sign), and (d) wavelength of the wave. (e) Find the maximum transverse speed of a particle in the string.

54. Two waves are described by

$$y_1 = 0.30\sin\left[\pi\left(5x - 200t\right)\right]$$

$$y_1 = 0.30\sin\left[\pi\left(5x - 200t + \frac{\pi}{3}\right)\right]$$

where y_1, y_2, and x are in meters and t is in seconds. When these two waves are combined, a traveling wave is produced. What are the (a) amplitude, (b) wave speed, and (c) wavelength of that traveling wave?

55. Four waves are to be sent along the same string, in the same direction:

$$y_1(x,t) = (4.00\,mm)\sin(2\pi x - 400\pi t)$$
$$y_2(x,t) = (4.00\,mm)\sin(2\pi x - 400\pi t + 0.7\pi)$$
$$y_3(x,t) = (4.00\,mm)\sin(2\pi x - 400\pi t + \pi)$$
$$y_2(x,t) = (4.00\,mm)\sin(2\pi x - 400\pi t + 1.7\pi)$$

What is the amplitude of the resultant wave?

56. The position of a particle is given by the expression $x = 4.00 \cos(3.00\pi t + \pi)$, where x is in meters and t is in seconds. Determine (a) the frequency and (b) period of the motion, (c) the amplitude of the motion, (d) the phase constant, and (e) the position of the particle at $t = 0.250$ s.

57. A 7.00-kg object is hung from the bottom end of a vertical spring fastened to an overhead beam. The object is set into vertical oscillations having a period of 2.60 s. Find the force constant of the spring.

58. The wave function for a traveling wave on a taut string is (in SI units) $y(x, t) = 0.350 \sin(10\pi t - 3\pi x + \pi/4)$ (a) What are the speed and direction of travel of the wave? (b) What is the vertical position of an element of the string at $t = 0$, $x = 0.100$ m? What are (c) the wavelength and (d) the frequency of the wave? (e) What is the maximum transverse speed of an element of the string?

59. A wave is described by $y = 0.020\ 0 \sin(kx - \omega t)$, where $k = 2.11$ rad m, $\omega = 3.62$ rad s, x and y are in meters, and t is in seconds. Determine (a) the amplitude, (b) the wavelength, (c) the frequency, and (d) the speed of the wave.

60. (a) Write the expression for y as a function of x and t in SI units for a sinusoidal wave traveling along a rope in the negative x direction with the following characteristics: A = 8.00 cm, λ = 80.0 cm, f = 3.00 Hz, and y (0, t) = 0 at t = 0. (b) What If ? Write the expression for y as a function of x and t for the wave in part (a) assuming that y (x, 0) = 0 at the point x = 10.0 cm.